with the help of the piston pin boss bearing and arrange to be movable within the pin.

5.3.8.1(c) Full floating type:

In this type the connection method of piston pin is designed in a simple way with no much difficulty. During the engine operation for avoiding damage on cylinder wall by the rubbing of piston pin, two circlips were attached with the piston pin boss. In this type, the piston pin is connected at the small end of the connecting rod and piston pin boss and allowed to float (move) easily with in the boss. Hence it is called as full floating type.

5.3.9 Crank Shaft

It converts the reciprocating motion of the piston into useful rotary motion of the output shaft. The crankshaft is enclosed in a crankcase and it is made up of Cast steel.

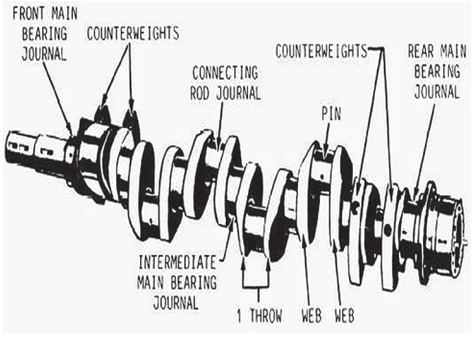


Figure 5.3.9(a) Crank Shaft

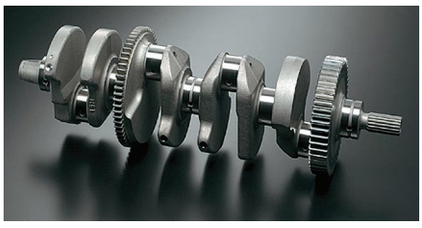


Figure 5.3.9(b) Crank Shaft

5.3.10 Vibration Damper

The power impulses of an engine result in torsional vibration in the crankshaft. If this torsional vibration were not reduced, the crankshaft might break. To avoid this, a vibration damper is mounted on the front of the crankshaft and it controls this vibration. Also a pulley is attached to the vibration damper to drive the fan.



Figure 5.3.10(a) Vibration Damper

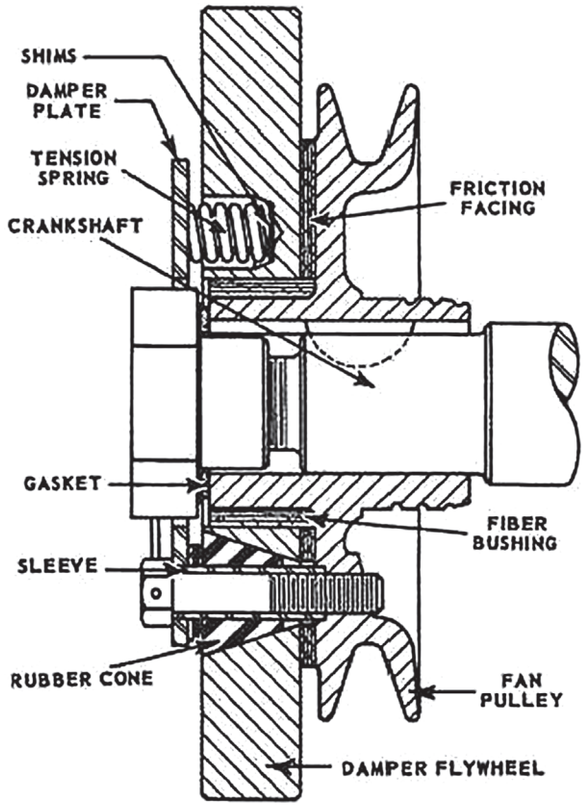


Figure 5.3.10(b) Schematic of Vibration Damper

5.3.11 Timing Gear

Timing gear is used to synchroinse the crankshaft and camshaft. The number of

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